

How Let's Talk Finance 1 is Organized

The 20 topics in **Let's Talk Finance 1** are organized around four ways that finance shapes public life beyond the household — the systems, institutions and rules that govern money at scale.

Part 1: Money, Payments and Financial Technology

How new forms of money and payment technology are changing, and how regulators are responding.

Part 2: Markets, Institutions and Financial Stability

How markets, credit ratings, insurers and banks are supervised, and what happens when the system comes under stress.

Part 3: Capital, Investment and Public Policy

How large flows of capital — green finance, infrastructure, trade, pensions, recovery spending — are directed and governed.

Part 4: Finance, Fairness and Global Cooperation

How financial systems address fairness, accountability and cooperation across borders.

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Part 1

Money, Payments and Financial Technology

Money, Payments and Financial Technology looks at what happens when a payment system changes faster than the rules built to govern it. Cryptocurrency moved from a niche curiosity to a market large enough that governments in Japan, the European Union and the United States each felt compelled to write dedicated rules, though they chose very different approaches. Central bank digital currencies raise a related but distinct question: not whether private money needs supervision, but whether central banks should offer a public digital alternative at all, and almost every country studying one has so far chosen not to launch.

The same tension between innovation and oversight runs through fintech regulation, where regulatory sandboxes let new payment models prove themselves under supervision before rules are finalised. It also runs through data privacy, where the same personal information that makes a service convenient becomes a liability the moment it crosses a border without adequate protection, and through financial inclusion, where mobile money and agent banking have brought hundreds of millions of previously unbanked adults into the formal financial system while a fully banked country like Japan faces a very different inclusion problem: keeping cash and branch access available as both quietly disappear. Across all five topics, the same question recurs: does a new technology need its own new rulebook, or can it be supervised inside the old one?

1.1. Cryptocurrency Regulation

1.1.1. Goal

Explain why governments have moved to regulate crypto-assets, and how the main approaches around the world differ.

1.1.2. Reading

Cryptocurrencies are digital assets recorded on a shared database called a blockchain. They can be sent between people without a bank in the middle. That design has advantages, but it also makes some old problems harder to control. Without a bank, it is harder to check who is sending money, harder to stop fraud, and harder to protect people who do not understand the risk.

For years, most governments watched from a distance. Two kinds of event changed that. The first was theft. In 2014 the Tokyo exchange Mt. Gox collapsed after losing a large amount of customer bitcoin, and in 2018 another Japanese exchange, Coincheck, lost about US\$530 million of a crypto-asset in a hack^[1]. The second was collapse. In November 2022, FTX, then one of the world's largest exchanges, failed in a few days with at least US\$8 billion of customer money missing; its founder was later sentenced to 25 years in prison for fraud^[2]. In both kinds of case, firms that looked established turned out to be poorly supervised.

Governments have responded in different ways. Japan acted early: from 2017, a company running a crypto exchange has had to register with the Financial Services Agency and follow rules on customer protection^[1]. The European Union chose a single large framework. Its Markets in Crypto-Assets Regulation, known as MiCA, applies across all member states; its rules for stablecoins started on 30 June 2024 and its rules for exchanges on 30 December 2024^[3]. The United States moved later and more narrowly: its first federal crypto

law, the GENIUS Act, was signed in July 2025 and covers only payment stablecoins^[4].

A stablecoin is a crypto-asset meant to hold a fixed value, usually equal to one dollar or one euro. All three approaches treat stablecoins carefully. Under MiCA and the GENIUS Act, an issuer must hold safe assets equal to every coin it puts into circulation and publish regular reports^{[3][4]}. In Japan, only a licensed bank, a registered fund-transfer firm, or a trust company may issue one for the public^[1]. The shared worry is that a coin advertised as “always worth one dollar” could fail if the assets behind it are not real or not safe.

The hard part is timing and balance. Rules that are too strict, or that arrive too late, can push activity into countries with weaker supervision. Rules that are too loose leave customers exposed, as Mt. Gox, Coincheck and FTX all showed. Because crypto moves easily across borders, regulators increasingly compare notes, even though their laws still differ.

1.1.3. Vocabulary Focus

cryptocurrency, blockchain, crypto exchange, stablecoin, issuer, supervision

1.1.4. Reading Questions

1. Why does the design of cryptocurrencies make some old problems harder to control?
2. What are the “two kinds of event” that the reading says changed how governments act?
3. How does the EU’s approach to crypto regulation differ from the United States’ approach?
4. What must a stablecoin issuer do under both MiCA and the GENIUS Act?
5. Why does the reading say regulators “increasingly compare notes”?

1.1.5. Discussion Questions

1. Japan regulated crypto exchanges years before the EU or the US. What are the advantages and disadvantages of being an early mover on rules like these?
2. The EU chose one law for all its members; the US chose a narrow law about stablecoins only. Which approach do you think is better, and why?
3. Should a stablecoin be treated like money, like an investment, or like something new? Explain your choice.
4. A friend says crypto does not need rules because “people choose the risk themselves.” How would you respond?
5. Advise a country that is deciding whether to copy MiCA, write a narrower law, or wait. What should it consider?

1.1.6. Source Notes

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1.2. Central Bank Digital Currencies

1.2.1. Goal

Explain what a central bank digital currency is, why most central banks are studying one, and why almost none have launched.

1.2.2. Reading

More than nine in ten of the world's central banks are working on a digital version of their national money. In a 2024 survey by the Bank for International Settlements, 85 of the 93 central banks that replied said they were exploring one^[1]. Yet by 2026 only three small economies had actually launched one for everyday public use, and use in all three has been very low.

A central bank digital currency, or CBDC, is official money issued in digital form by the central bank itself. The cash in your pocket is already a claim on the central bank. The money in your bank account, by contrast, is a claim on a commercial bank. A “retail” CBDC would be central-bank money that ordinary people could hold directly, without a bank in between, and use for daily payments. A separate “wholesale” type, used only for settlement between financial institutions, is less controversial and further along.

The launches so far are small and disappointing. The Bahamas issued the world's first, the Sand Dollar, in October 2020. Jamaica followed in 2022, and Nigeria launched the eNaira in 2021, the first in Africa^[5]. Take-up has been weak in every case. The International Monetary Fund found that the large majority of eNaira digital wallets had never been used at all^[2].

The larger economies are still testing. China has run by far the biggest trial. Its digital yuan had handled a cumulative 7 trillion yuan, about US\$1 trillion, in transactions by mid-2024, across 17 regions^[3]. But much of that use was pushed by the

government, through things like staff payments, rather than chosen by shoppers, who already rely on Alipay and WeChat Pay. The European Central Bank ran a “preparation phase” from 2023 and moved to the next stage in October 2025^[6]. Even so, a digital euro cannot be issued unless EU lawmakers pass a new law, and a first issuance would not come before 2029^[4]. The Bank of Japan has run a pilot since 2023 and says no decision to issue has been made^[7]. A government inquiry into Sweden's e-krona concluded there was not yet enough need for one^[8].

The reasons given for the work are mostly defensive. Central banks want to keep a public form of money available as cash use falls, to make payments faster and cheaper, and to reduce dependence on a few large private payment firms and on privately issued stablecoins. The reasons for caution are just as practical: people might pull money out of banks in a crisis, a digital record raises privacy questions, and the public has so far shown little appetite for it.

1.2.3. Vocabulary Focus

central bank digital currency (CBDC), retail CBDC, wholesale CBDC, claim (on the central bank), pilot, disintermediation

1.2.4. Reading Questions

1. What is the difference between the cash in your pocket and the money in your bank account, according to the reading?
2. What is a “retail” CBDC, and how does it differ from a “wholesale” one?
3. What does the reading say happened with Nigeria’s eNaira after it launched?
4. Why is China’s 7 trillion yuan figure described with some caution?
5. What conditions must be met before the European Central Bank can issue a digital euro?

1.2.5. Discussion Questions

1. The reading says central banks give “defensive” reasons for this work. Are those reasons strong enough to justify the cost and effort? Give your view.
2. Compare the risks of issuing a CBDC with the risks of not issuing one. Which set of risks is more serious?
3. If a digital yen were introduced in Japan, what would make you use it instead of cash or a payment app? What would put you off?
4. Several countries launched a CBDC and then saw very low use. What lessons should other central banks take from that?
5. Advise a central bank that is deciding between launching a CBDC now, running a longer pilot, or stopping the project. What should it weigh?

1.2.6. Source Notes

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1.3. Financial Technology (Fintech) Regulation

1.3.1. Goal

Compare the main ways governments regulate fintech, and show why a firm licensed in one country usually cannot operate in another.

1.3.2. Reading

“Fintech” means financial services delivered mainly through software: app-based payments, online lending, digital-only banks. As these firms grew, governments had to decide how to supervise them. In 2016 the United Kingdom’s Financial Conduct Authority tried a new answer, the world’s first “regulatory sandbox”^[1]. A sandbox lets a new firm test a product with real customers for a limited time, under tailored rules and close supervision. The idea spread fast. By 2020 the World Bank counted 73 sandboxes in 57 countries^[2]. But a sandbox is only one tool, and countries have taken very different overall approaches.

The European Union chose broad, EU-wide law. Its Payment Services Directive, known as PSD2, took effect in 2018. It created the legal basis for “open banking” by requiring banks to let licensed third parties reach customer payment accounts, usually through a standard software connection called an API^[3]. A firm authorised in one EU country can then offer services across all the others. The EU agreed a further update, PSD3, in late 2025.

The United States has no single fintech law or regulator. A money-transfer firm must be licensed separately in each state where it operates. A firm serving the whole country may therefore need dozens of licences, which raises the cost of launching and tends to favour large, well-funded companies. Attempts to write one national open-banking rule have so far stalled in the courts.

India built public infrastructure instead of a broad rulebook. Its Unified Pay-

ments Interface, launched in 2016 and overseen by the Reserve Bank of India, processed more than 240 billion transactions in the 2025–26 financial year^[4]. That makes it the world’s largest real-time payment system, and it is free to use for individuals. Private apps compete on top of it, but the underlying rails are public, so no single company controls access.

Japan’s Financial Services Agency uses a lighter, cooperation-based model. A 2017 amendment to the Banking Act, in force from 2018, created a registration system for the payment and data firms that connect to banks, and required banks to make best efforts to build open-API systems for them within two years^[5]. A 2020 change then split non-bank money-transfer providers into three licence tiers by transfer size. Only the tier for the largest transfers needs full FSA authorisation; the two smaller tiers need registration only^[6].

Because each of these frameworks is written into national law, they do not travel. A payment firm licensed in London, Frankfurt, Mumbai or Tokyo must usually get a fresh authorisation before serving customers elsewhere. That is why fintech firms expand country by country, not all at once, and why a service that is normal in one market can be unavailable in the next.

1.3.3. Vocabulary Focus

fintech, regulatory sandbox, open banking, application programming interface (API), licensing, real-time payment system

1.3.4. Reading Questions

1. What does a regulatory sandbox allow a new firm to do?
2. How does an EU authorisation help a fintech firm, according to the reading?
3. Why might a fintech firm need “dozens of separate licences” to operate across the United States?
4. What did India build instead of a broad fintech law, and how large has it become?
5. What does the reading say a payment firm must usually do before serving customers in another country?

1.3.5. Discussion Questions

1. The EU wrote one broad law; the United States relies on many state licences. What are the advantages and disadvantages of each approach?
2. India focused on building a public payment system rather than regulating private ones. When is that a better choice for a government?
3. A regulatory sandbox gives a few firms relaxed rules for a while. Is that fair to firms not chosen for the sandbox? Explain.
4. Japan’s model relies on banks and fintech firms agreeing terms with each other. What could go wrong with that, and what could go right?
5. Advise a government that wants to encourage fintech but is worried about consumer losses. Which two rules would you tell it to put in place first?

1.3.6. Source Notes

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1.4. Data Privacy and Protection

1.4.1. Goal

Explain how personal financial data is protected by law, and why moving that data between countries is the hardest part.

1.4.2. Reading

In 2023 an Irish regulator fined the technology company Meta EUR 1.2 billion. It was the largest privacy fine ever imposed in the European Union^[1]. Meta's mistake was not losing the data or selling it. It was moving Facebook users' personal information from Europe to the United States without enough legal protection against access by US authorities. The case shows what modern data-privacy law is mainly about. It is not just keeping data safe. It is controlling where the data goes and who can reach it.

Financial firms hold large amounts of sensitive personal data: account numbers, transaction histories, credit records, copies of identity documents. They want to use it, for credit scoring, fraud checks and new services. Privacy law sets the limits on how they can. It usually requires a clear reason for collecting each piece of data, the person's consent, and a duty to keep only what is needed.

The European Union's rules, known as the GDPR, have applied since 2018 and let regulators impose very large fines. By early 2025, EU regulators had issued close to EUR 6 billion in GDPR fines in total, and companies were reporting hundreds of data breaches every day^[2]. Other regions rely more on penalties after a specific breach. After a 2017 attack on the US credit bureau Equifax exposed the records of about 147 million people, the company agreed to pay up to US\$700 million to settle claims from US regulators and states^[3]. The United States still has no single national privacy law. California and

several other states have written their own.

Japan was an early mover. Its Act on the Protection of Personal Information dates from 2003. Updates in 2017 and 2022 strengthened individual rights and made serious data breaches reportable to the Personal Information Protection Commission^[6]. In 2019 the EU and Japan recognised each other's systems as offering "adequate" protection^[4]. Personal data can now flow both ways without extra paperwork. It was the first such mutual arrangement the EU concluded^[7]. Newer laws elsewhere, such as India's Digital Personal Data Protection Act of 2023, follow similar lines^[5].

The unresolved problem is cross-border transfer. Data moves around the world in seconds, but privacy rules stop at national borders. A company can be careful and fully lawful at home and still break the law by sending the same data to a country whose protections, or whose government surveillance powers, are judged too weak. That is exactly what the Meta fine was about. For a bank or payment firm that runs its systems across several countries, deciding where customer data may legally be stored and processed has become a routine part of the job.

1.4.3. Vocabulary Focus

personal data, data breach, data protection authority, consent, cross-border data transfer, adequacy decision

1.4.4. Reading Questions

1. What exactly did Meta do wrong, according to the reading?
2. Why do financial firms want access to large amounts of personal data?
3. How does the United States protect financial data differently from the European Union?
4. What did the EU and Japan agree in 2019, and why does it matter?
5. What does the reading mean when it says “privacy rules stop at national borders”?

1.4.5. Discussion Questions

1. The reading describes a tension between using data and protecting it. Where would you draw the line for a bank that wants to use customer data to detect fraud?
2. Is a very large fine, like the one imposed on Meta, an effective way to change how big companies handle data? Explain.
3. Compare protecting data through strict rules (the EU approach) with protecting it through breach penalties (the US approach). Which is likely to work better, and for whom?
4. Should a person be able to refuse to let their bank share their data, even if that means fewer services or a worse credit offer? Give your view.
5. Advise two countries that want to allow data to flow freely between them. What would each need to check about the other first?

1.4.6. Source Notes

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1.5. Financial Inclusion Initiatives

1.5.1. Goal

Explain what financial inclusion means, how developing countries have widened access to accounts, and why it means something different in a rich country.

1.5.2. Reading

About 1.3 billion adults worldwide still have no bank account of any kind. That is far fewer than a decade ago. The share of adults with an account rose from roughly half in 2011 to nearly 80% by 2024^[1]. But progress has been uneven, and most of the people still left out live in just eight countries.

“Financial inclusion” means giving people who are shut out of the formal system practical access to an account, payments, savings and credit. In lower-income countries the main tools have been mobile technology and public infrastructure.

Mobile money has been the biggest change in Sub-Saharan Africa. M-Pesa was launched by the telecoms company Safaricom in Kenya in 2007. It lets people hold and send money using a basic phone, with cash paid in and out through local shop agents rather than bank branches. Sub-Saharan Africa is now the only region where more adults have a mobile money account than a bank account^[1]. In Kenya itself, the share of adults with access to formal financial services has reached about 85%^[2].

Governments have also built the infrastructure directly. India’s Jan Dhan programme, started in 2014, had opened more than 500 million basic bank accounts by the mid-2020s^[5]. Most are held by women and rural residents. In Brazil, the central bank acted as regulator and infrastructure builder at once, launching an instant-payment system called Pix in 2020. Within about a year, two-thirds of Brazilian adults were using it, including mil-

lions who had never made an electronic transfer before^[3].

Having an account, though, is not the same as using one. In India, about 16% of account holders have an inactive account — roughly four times the rate in other middle-income countries^[1]. A programme can put a card in someone’s hand and still not change how they manage money.

In a rich country the problem is different again. Japan is effectively fully banked, so inclusion is less about opening a first account. It is more about how people pay and who is left behind as things change. Japan has stayed unusually cash-reliant: the government reported that the cashless share of consumer spending reached about 43% in 2024, meeting its 40% target a year early^[4]. The concerns are for elderly and rural customers as bank branches and cash machines close, and for foreign residents who face language and paperwork barriers to opening an account. The historical inclusion tool was the post office. Japan Post Bank, with roots in a savings system from 1875 and around 24,000 branches, long made sure that even remote areas had somewhere to save^[6]. Whether the goal is a first account in Kenya or keeping cash access in rural Japan, the harder part is not reaching people once but making the service one they will actually use.

1.5.3. Vocabulary Focus

financial inclusion, the unbanked, mobile money, agent banking, instant payment system, cashless payment

1.5.4. Reading Questions

1. How much did the share of adults worldwide with an account change between 2011 and 2024?
2. How does M-Pesa let people hold and move money without a bank branch?
3. What did India's Jan Dhan programme do, and who mainly holds the accounts?
4. Why does the reading say financial inclusion "means something different" in Japan?
5. What role did Japan Post Bank historically play in financial inclusion?

1.5.5. Discussion Questions

1. In several countries, governments rather than private banks built the tools that widened access. When is that the right choice, and when is it not?
2. Having an account is not the same as using one. What might stop someone from using an account they have been given?
3. Compare Kenya's mobile-money route with Brazil's central-bank payment system. What are the strengths of each?
4. In a rich country, who is most at risk of being "left behind" as cash and branches disappear, and what could be done for them?
5. Advise a government that wants to bring its unbanked population into the financial system. What two steps would you tell it to take first?

1.5.6. Source Notes

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Part 2

Markets, Institutions and Financial Stability

Markets, Institutions and Financial Stability turns to the machinery that is supposed to catch trouble before it spreads. Circuit breakers and daily price limits exist to slow a market panic down long enough for information, not fear, to set prices again, while credit rating agencies translate a borrower's complex financial position into a single letter grade that investors around the world rely on, sometimes too heavily. Insurance regulation asks a related question over a much longer time horizon: whether an insurer holds enough capital today to make good on promises it will not have to pay out for decades.

Stress testing brings these ideas together directly, deliberately imagining a bank's worst plausible future to check whether its capital buffer would survive it, a test that only measures the risk it is designed to measure, as Signature Bank's failure showed. Sovereign debt management closes the part by moving from private institutions to governments themselves, examining how a country manages, restructures and, when necessary, defaults on debt owed not to one lender but to thousands of bondholders. Each of these five topics is really the same argument told through a different institution: a financial system is only as stable as the assumptions behind the safeguards nobody hopes to actually use.

2.1. Regulatory Responses to Market Volatility

2.1.1. Goal

Explain how exchanges and regulators try to slow a falling market, using circuit breakers and trading halts, and why these tools are debated.

2.1.2. Reading

On 5 August 2024 the Nikkei 225, Japan's main stock index, fell 4,451 points in a single day, a drop of 12.4% and the largest one-day points fall in its history^[1]. Futures trading on the Osaka Exchange was halted for a short period when prices moved past a set limit. Markets in South Korea paused the same day. A day later, share prices in Tokyo recovered much of the loss. Sharp swings like this are what regulators call market volatility, and over the past forty years they have built tools to manage it.

The best known tool is the circuit breaker, a rule that stops trading for a fixed time when prices fall by a set amount. The idea came from the United States after 19 October 1987, a day known as “Black Monday”, when the Dow Jones index lost 22.6% in one session. A government report, the Brady Commission report, recommended coordinated pauses so that buyers and sellers could catch up with the news^[2]. The exchanges adopted matching rules the following year. Today the main United States market has three levels, set at falls of 7%, 13% and 20% in the S&P 500 index. The first two trigger a 15-minute halt; the third closes the market for the rest of the day. Before 2020 the mechanism had been used only once, in 1997. Then the 7% level was reached four times in a single fortnight in March 2020, during the first weeks of the pandemic^[3].

Other markets use their own versions. After the “flash crash” of 6 May 2010, when United States prices fell and rebounded within minutes, regulators added limits that pause trading in a single

stock if its price moves too far too fast. The European Union allowed national regulators to ban the short selling of some shares for several weeks in March 2020. Exchanges in Tokyo, Seoul, Shanghai and elsewhere in Asia set daily price limits on individual stocks, so a share can only rise or fall by a capped percentage before trading in it stops.

Not every attempt has worked. China introduced a market-wide circuit breaker in January 2016 and removed it after four days. Traders rushed to sell before the halt could take effect, which made the fall worse; on the fourth day the market closed just half an hour after opening. This points to the central argument against the tools, sometimes called the magnet effect: a pause meant to calm a market can instead pull prices toward the trigger as people race to trade first. Supporters reply that a short break gives firms time to check their systems and gives regulators time to spot errors, and that a planned stop is safer than a disorderly one. After more than thirty years the research is still mixed. Most large exchanges have kept their circuit breakers, while continuing to adjust the levels and the timing in the light of each new episode.

2.1.3. Vocabulary Focus

market volatility, circuit breaker, trading halt, stock index, short selling, daily price limit

2.1.4. Reading Questions

1. What happened to the Nikkei 225 on 5 August 2024, and how did the market move the next day?
2. What is a circuit breaker, and what event led the United States to create one?
3. What are the three levels used in the main United States market, and what does each one do?
4. What change did regulators make after the “flash crash” of May 2010?
5. Why did China remove its market-wide circuit breaker after four days?

2.1.5. Discussion Questions

1. A circuit breaker forces a pause on people who may want to trade. When is it fair to stop a market, and when is it not?
2. The reading gives one case where a circuit breaker seemed to make a fall worse. Does that mean the tool is a bad idea? Explain your view.
3. Daily price limits on single stocks are common in Asia but rare in the United States. What might explain that difference?
4. Some investors say pauses just delay bad news. Others say a break prevents panic. Which argument do you find stronger, and why?
5. You advise an exchange that has never had a circuit breaker. Recommend whether it should adopt one, and set out the two main risks it should weigh.

2.1.6. Source Notes

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2.2. Credit Rating Agencies Oversight

2.2.1. Goal

Explain what a credit rating is, why the way rating agencies are paid creates a conflict of interest, and how regulators have responded since 2008.

2.2.2. Reading

It is easy to think of a credit rating as a fact, like a temperature reading. It is not. A credit rating is an opinion about how likely a borrower is to repay. It is produced by a private company and expressed on a letter scale, from the top grade of AAA down to grades that signal a high risk of default, meaning a failure to pay on time. Governments, companies and bundles of loans all carry ratings. The opinions matter because many large investors, by law or by their own rules, may only hold bonds above a certain grade. A downgrade can therefore force selling and raise a borrower's cost of new debt.

Three firms produce the great majority of these opinions. Standard & Poor's, Moody's and Fitch together accounted for about 94% of all outstanding ratings in the United States at the end of 2023^[1]. All three are paid by the borrower whose bond they rate. This is the conflict of interest at the centre of the industry: the agency's customer is the party that wants a high grade, and that party can take its business to a competitor. Before 2008, agencies gave top grades to large volumes of United States mortgage-backed bonds that later collapsed in value, which helped turn a housing downturn into a global crisis.

Regulators responded on both sides of the Atlantic. In the United States, the Securities and Exchange Commission was given clearer authority to oversee registered rating agencies, and in 2015 Standard & Poor's paid about US\$1.375 billion to settle government claims over its pre-crisis mortgage ratings^[2]. In the European Union, a 2009 regulation required agen-

cies to register. Since 2011 the European Securities and Markets Authority has been their single supervisor, with the power to fine them, and several firms have paid penalties for control failures^[3]. The rules also require agencies to publish their methods and their past accuracy, so that users can judge them.

Sovereign ratings, meaning ratings of national governments, show why the opinions still carry weight. When Standard & Poor's lowered the United States from AAA in 2011, and when Fitch and Moody's followed in later years, the decisions drew sharp official criticism but still moved markets^[4]. France was downgraded by more than one agency in 2025^[5]. Japan takes a slightly different path. Alongside the global firms, it has two home-grown agencies, Japan Credit Rating Agency and Rating and Investment Information. All agencies that operate in Japan must register with the Financial Services Agency under a regime in force since 2010^[6]. Oversight has not removed the basic tension in the business model. It has instead tried to make the opinions more transparent, more comparable and easier to challenge, so that investors treat a rating as one view among several rather than a guarantee.

2.2.3. Vocabulary Focus

credit rating, credit rating agency, default, conflict of interest, sovereign rating, investment grade

2.2.4. Reading Questions

1. Why does the reading say a credit rating is an opinion rather than a fact?
2. What share of United States ratings do the three largest agencies produce, and how are those agencies paid?
3. What did rating agencies do before 2008 that made the financial crisis worse?
4. What did Standard & Poor's agree to in 2015, and why?
5. How does oversight of rating agencies in the European Union work, and who is the supervisor?

2.2.5. Discussion Questions

1. The borrower pays the agency that rates it. What other ways of paying for ratings can you think of, and what would go wrong with each?
2. Some investors are required to hold only highly rated bonds. Is that rule sensible, or does it give the agencies too much power? Explain.
3. Governments have criticised agencies for lowering their ratings. Should a private firm be able to judge a national government's debt? Give reasons.
4. Japan supports its own rating agencies alongside the global firms. What are the advantages and drawbacks of having national agencies?
5. You advise a regulator that wants investors to rely less on ratings. Suggest two practical steps it could take.

2.2.6. Source Notes

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2.3. Insurance Market Regulation

2.3.1. Goal

Explain why insurers are regulated differently from banks, what a solvency rule is meant to do, and how the main systems compare.

2.3.2. Reading

In April 1997 Nissan Mutual Life, a mid-sized Japanese insurer, was ordered to stop selling policies because it could not meet its future promises to policyholders. It was the first failure of a Japanese life insurer since the Second World War. Several more followed between 2000 and 2001, among them Chiyoda Life and Kyoei Life^[1]. An insurer takes in premiums today and pays claims that may fall due decades later. Regulation of the industry is built around one question: will the company still be able to pay when the claim finally comes?

The tool for answering that question is a solvency rule. It requires an insurer to hold capital, meaning assets worth more than its expected future claims, so that it can absorb a bad year without failing. The European Union's system, known as Solvency II and in force since 2016, sets the required capital at the level needed to survive a loss expected only once in 200 years^[2]. Japan's Financial Services Agency has used a solvency margin ratio since the late 1990s. A reading below 200% triggers official intervention, and the agency is moving to a newer, more risk-sensitive standard for the 2026 financial year^[1].

The United States regulates insurance mainly at the state level rather than nationally. State regulators apply a shared formula, called risk-based capital, developed by the National Association of Insurance Commissioners. The formula sets action points where a regulator must step in as an insurer's capital falls^[3]. If an insurer does fail, state guaranty associations, funded by the industry, cover policy-

holder claims up to set limits, such as US\$300,000 for a death benefit in many states^[4]. The United Kingdom kept a version of the European rules after leaving the bloc and has since adjusted them under the name Solvency UK.

International bodies have tried to add a layer above the national systems. In 2008 the American insurer AIG came close to collapse and needed roughly US\$180 billion of government support. The losses came from contracts written by one small division, and the episode showed that a large insurance group can threaten the wider financial system. In response, the International Association of Insurance Supervisors, a body whose members cover almost all of the world's insurance premiums, agreed a global Insurance Capital Standard in 2024 for the biggest cross-border groups^[5]. It is meant to give supervisors in different countries a common measure for the same group.

The common thread across all these systems is patience. Unlike a bank run, an insurer's trouble usually builds slowly, over years of underpricing risk or earning too little on its investments, and solvency rules are designed to catch that drift early. The Japanese failures around 2000 came after a long period of guaranteeing returns that the insurers' own investments could no longer earn.

2.3.3. Vocabulary Focus

insurer, premium, claim, solvency rule, capital, policyholder

2.3.4. Reading Questions

1. What happened to Nissan Mutual Life in 1997, and why was it significant?
2. What is a solvency rule meant to ensure?
3. How does the European Union's Solvency II decide how much capital an insurer must hold?
4. How is insurance regulated in the United States, and what happens to policyholders if an insurer fails?
5. What did the near-collapse of AIG in 2008 show, and how did international regulators respond?

2.3.5. Discussion Questions

1. The reading says an insurer's trouble usually builds slowly, unlike a bank run. Why might slow trouble be harder for a regulator to act on, not easier?
2. The United States regulates insurance state by state; the European Union uses one shared system. What are the strengths and weaknesses of each approach?
3. Guaranty associations repay failed insurers' customers up to a limit. Who should pay for that protection, and should there be a limit at all?
4. Should the largest insurance groups face an extra global standard on top of national rules? Give reasons for and against.
5. You advise a regulator worried that insurers are earning too little on safe investments. Suggest two things it could watch for as warning signs.

2.3.6. Source Notes

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2.4. Stress Testing and Risk Management

2.4.1. Goal

Explain what a bank stress test is, show one case where it worked and one risk it missed, and describe how the practice is used today.

2.4.2. Reading

Can a bank survive a bad year on paper before it has to survive one in real life? That is the question a stress test tries to answer. A regulator sets out a severe but plausible scenario, such as a deep recession with rising unemployment and falling house prices. Each large bank must then estimate its losses under that scenario and show how much of its capital buffer would remain. The capital buffer is the cushion of shareholders' money that absorbs losses before depositors are at risk. If the buffer falls too low, the bank must raise more capital or hold back dividends.

The practice was proven in the United States in 2009. In the depths of the financial crisis, regulators tested 19 large banks, published the results, and found that ten needed about US\$75 billion of new capital between them^[1]. Because the method and the numbers were made public, investors could see which banks were sound, and confidence returned faster than expected. The test became an annual fixture, and the European Union, the United Kingdom and Japan built their own versions. In the 2025 European Union exercise, 64 banks were tested against a scenario that wiped out roughly 370 basis points, or 3.7 percentage points, of their capital ratios on average^[2].

The limits of the tool became clear in March 2023, when Signature Bank failed in the United States. The bank held about US\$110 billion in assets, well below the US\$250 billion threshold above which the annual stress test was mandatory^[3]. Its problem was not a shock the recent sce-

narios had focused on; it was contagion, a sudden loss of confidence after a similarly sized rival collapsed two days earlier. On 10 March, Signature lost about a fifth of its total deposits in a matter of hours, faster than any historical run the tests had modelled. A test only measures the dangers it is designed to measure.

Regulators have drawn two lessons. The first is to vary the scenarios, so that a test one year might centre on a property crash and the next on a sudden jump in interest rates or a technology shock. The 2022 European climate test was described by its own authors as a learning exercise rather than a pass-or-fail check^[5]. The second lesson is to treat the test as one instrument among several, alongside day-to-day supervision and the bank's own internal risk models. In Japan, the Bank of Japan and the Financial Services Agency run joint tests covering hundreds of institutions and publish the results in a twice-yearly report on the financial system^[4]. The exercise is now a routine part of oversight in every major economy, valued less as a precise forecast than as a structured way to ask what could go wrong.

2.4.3. Vocabulary Focus

stress test, scenario, capital buffer, recession, basis point, risk model

2.4.4. Reading Questions

1. What does a stress test require a bank to estimate, and what must the bank do if its buffer falls too low?
2. Why did the 2009 United States stress test help restore confidence?
3. Why was Signature Bank not part of the annual United States stress test?
4. What kind of shock caused Signature Bank's losses, and how fast did depositors withdraw money?
5. What two lessons have regulators drawn from that failure?

2.4.5. Discussion Questions

1. The 2009 test worked partly because the results were published. What are the risks of making each bank's results public, and do they outweigh the benefits?
2. A stress test only measures the dangers it is designed to measure. How can a regulator guard against missing the next new risk?
3. Should smaller banks face the same tests as the largest ones? Give reasons for and against.
4. The reading says a stress test is valued "less as a precise forecast than as a structured way to ask what could go wrong." Is that a useful thing for regulators to spend money on? Explain.
5. You advise a regulator designing next year's scenario. Suggest two shocks it should test for, and say why you chose them.

2.4.6. Source Notes

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2.5. Sovereign Debt Management

2.5.1. Goal

Explain how a national government borrows, why it matters who holds the debt and in which currency, and what happens when a government cannot pay.

2.5.2. Reading

Japan's central government owes more than twice the value of everything the country produces in a year. By the International Monetary Fund's 2026 figures, Japan's general government gross debt is about 206% of national output, by far the highest of any large advanced economy^[1]. Older estimates put the figure higher still, above 230%. Yet Japan has no debt crisis, low borrowing costs and no sign of panic. The reason is who lends the money. As of March 2026 the Bank of Japan alone held about 48% of all Japanese government bonds, and foreign investors held only about 8%; the rest sits with Japanese banks, insurers and pension funds^[2]. The debt is almost entirely domestic and in yen, so there is no foreign creditor to lose confidence and no foreign currency to find.

Most governments borrow the same basic way. A treasury or a specialist debt management office sells bonds, which are tradable promises to pay interest and repay a fixed sum on a set date, and short-term bills of under a year. It decides how much to sell, at what mix of maturities, and by what method, usually an auction open to competing bidders. Two choices shape the risk. The first is currency: borrowing in your own money removes the danger that a falling exchange rate makes the debt harder to repay. The second is maturity: spreading repayment dates over many years reduces the risk of having to refinance a large amount at a bad moment. The United Kingdom's Debt Management Office, for example, keeps

the average life of its bonds unusually long, at about 14 years^[3].

The United States shows a different balance. About 31% of United States federal debt held by the public is owned by foreign investors, with Japan the largest single holder, and the government paid roughly US\$282 billion in interest to those foreign holders in 2025^[4]. That reliance on outside lenders is manageable for a country whose currency the world wants to hold, but it would be dangerous for a smaller economy.

When debt genuinely cannot be paid, a government must restructure it. That means negotiating with creditors to stretch out repayment dates or reduce the amount owed, a reduction known as a haircut. Sri Lanka defaulted in 2022 and agreed a deal with bondholders in 2024 that cut the value of its bonds by roughly a quarter^[5]. Lower-income countries such as Zambia and Ghana have restructured through a shared process set up by the Group of Twenty, a body that brings together the world's largest economies. Zambia's 2024 deal wrote off about US\$840 million of what it owed^[6]. These cases share a feature Japan lacks: debt owed to foreigners, often in dollars, which is the hardest kind to carry when trouble comes.

2.5.3. Vocabulary Focus

sovereign debt, government bond, debt management office, bond auction, debt restructuring, haircut

2.5.4. Reading Questions

1. How large is Japan's government debt compared with national output, and why does the reading say there is no crisis?
2. Who holds most Japanese government bonds, and how much do foreign investors hold?
3. What are the two choices that shape the risk in how a government borrows?
4. How does the United States position differ from Japan's, and why is that manageable for the United States?
5. What does it mean to restructure debt, and what is a haircut?

2.5.5. Discussion Questions

1. Japan carries very high debt safely because it is owed at home in yen. Does that mean the level of debt does not matter? Explain your view.
2. Borrowing from foreign investors brings in money but adds risk. When is it worth it for a government, and when is it not?
3. A haircut means lenders are repaid less than they were promised. Is that fair to those lenders? Give reasons.
4. Lower-income countries often must borrow in dollars rather than their own currency. What could be done to change that?
5. You advise a government setting up a new debt management office. Suggest two rules it should follow from the start, and say why.

2.5.6. Source Notes

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Part 3

Capital, Investment and Public Policy

Capital, Investment and Public Policy examines how governments direct large, long-lived flows of money toward public goals rather than private returns alone. Green finance channels capital toward environmental projects through green bonds and an increasingly strict taxonomy of what counts as genuinely sustainable, while infrastructure investment asks a more basic question: how a country pays for the roads, ports and power grids that no single investor would build alone, often through public-private partnerships that split the risk between them. Trade policy and tariffs cover a third form of government intervention, one that raises prices for consumers as a deliberate tool of economic or political strategy rather than a market failure to correct.

Pension reform and the post-COVID economic recovery close the part by looking at how public policy manages long-run demographic and economic shocks: an ageing population that slowly outgrows a pension system designed for a younger one, and a global pandemic recovery that left some economies richer than before and others permanently behind. These five topics share a common thread: each is a case where markets alone would produce an outcome society has decided not to accept, and government policy is the tool used to change it.

3.1. Green Finance Initiatives

3.1.1. Goal

Explain what makes a green bond different from an ordinary bond, and why standard-setters and regulators have moved to define the “green” label.

3.1.2. Reading

Two companies borrow the same amount of money to build the same wind farm. The first sells an ordinary bond and can spend the money however it likes; no one checks afterwards. The second sells a green bond, and that label is a promise with rules attached. The company must spend the proceeds only on defined environmental projects, keep that money in a separate account, and report later on where it went and what it achieved^[1]. The wind farm is identical. What differs is the commitment about the use of proceeds and the disclosure that comes with it.

Green finance means money raised specifically for environmentally beneficial activity, mostly through green bonds. The European Investment Bank sold the first one in 2007, and the market grew slowly until common rules appeared^[2]. Since 2014 most issuers have followed the Green Bond Principles, a voluntary standard from the International Capital Market Association. It rests on four duties: define the eligible projects, explain how they are chosen, track the money separately, and report on both spending and results^[1]. By early 2025 the green bond market was worth about US\$2.9 trillion, with roughly US\$700 billion issued during 2024^[3].

Because “green” is otherwise a label an issuer picks for itself, regulators have begun to define it. The European Union’s Green Bond Standard has applied since December 2024. A bond may be called a “European Green Bond” only if at least 85% of the proceeds fund activities on the EU’s official taxonomy, its classified list of sustainable activities, with an external re-

view before and after issuance^[4]. The goal is to limit greenwashing, which means claiming an environmental benefit that the label does not deliver.

Japan has taken a distinctive step at government level. In February 2024 it became the first national government to sell climate transition bonds, in two parts of JPY 800 billion each, JPY 1.6 trillion together^[5]. The full programme aims to raise about JPY 20 trillion over roughly a decade to help heavy industry, power and transport cut emissions. Japan plans to repay the bonds partly from future carbon-pricing revenue, such as a fossil-fuel levy due from 2028, rather than from general taxes^[5]. Transition bonds are more debated than ordinary green bonds, because they fund high-polluting sectors that are trying to change rather than activities that are already clean.

Emerging economies use these tools too. Chile sold Latin America’s first sovereign green bond in 2019 and by late 2025 had about US\$56 billion of labelled sustainable debt outstanding, more than any other developing country^[6]. Across all these issuers, the bond itself is ordinary borrowing; what investors are paying for is the enforceable promise to spend and disclose in a set way. That is why the argument over green finance now centres on the standards and the outside checks, not on the wind farms and rail lines the money builds.

3.1.3. Vocabulary Focus

green bond, use of proceeds, taxonomy, greenwashing, transition bond, external review

3.1.4. Reading Questions

1. According to the reading, what is the difference between an ordinary bond and a green bond that fund the same project?
2. What four duties does an issuer take on under the Green Bond Principles?
3. How large was the green bond market by early 2025, and how much was issued during 2024?
4. Why have regulators such as the EU started to give “green” a legal definition?
5. Why does the reading say transition bonds are “more debated” than ordinary green bonds?

3.1.5. Discussion Questions

1. The reading says the wind farm is the same whether or not the bond is labelled green. In your view, what real difference does the green label make?
2. Is an external review before and after issuance enough to stop greenwashing? What more could be required?
3. Japan's transition bonds fund industries that still pollute heavily. Is that a fair use of climate finance, or a weakness in the idea? Explain.
4. Advise a company issuing its first green bond on three things it should do so that investors trust the label.
5. Should a country make the green-bond label a legal standard, as the EU has done, or keep it voluntary? Give your reasons.

3.1.6. Source Notes

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3.2. Infrastructure Investment

3.2.1. Goal

Explain the gap between the infrastructure the world needs and what it will pay for, and show who fills that gap besides national governments.

3.2.2. Reading

It is natural to think that roads, bridges, power grids and water systems are built and paid for by government. Governments plan and own most infrastructure, often raising the money through government bonds — but they have never been its only funders, and the shortfall between what the world invests and what it needs is now beyond public budgets alone. The Global Infrastructure Hub, set up by the Group of Twenty, estimates a need of about US\$97 trillion of infrastructure investment worldwide by 2040, against roughly US\$79 trillion likely to be spent — an infrastructure funding gap of around US\$18 trillion^[1].

The gap has two forms: capacity that is never built, and, more common in richer countries, old infrastructure that now needs expensive repair. In 2025 the American Society of Civil Engineers graded United States infrastructure at C, up from C-minus in 2021, but still identified a US\$3.7 trillion shortfall over ten years^[2]. A broader estimate puts the European Union's yearly gap on strategic investment — climate, digital and infrastructure combined — at about EUR 481 billion^[3].

Japan shows the ageing problem in sharp form. It built most of its expressways, bridges and tunnels between the 1950s and 1970s. By 2033 about 60% of its roughly 730,000 road bridges will be at least 50 years old, the point where damage spreads quickly and full repair becomes unavoidable^[4]. Japan still spends heavily on public works, but a rising share now goes to maintaining what exists, and

a shrinking workforce makes the repairs harder to staff.

So who fills the gap? One answer is private money through public-private partnerships, or PPPs, where a private company designs, builds, finances and operates a public asset for a fixed period, called a concession, earning tolls or fees before ownership returns to the state. Kenya's Nairobi Expressway, a 27-kilometre toll road opened in 2022, was built this way by a Chinese state contractor — Kenya's first toll expressway. The contractor paid about 80% of the roughly US\$668 million cost and collects tolls for about three decades before handing the road over^[5].

Multilateral development banks, lenders owned by many governments together, are another source. The World Bank Group committed US\$118.5 billion in its 2025 financial year and helped attract a further US\$69.9 billion of private capital^[6]. Much of this is blended finance: public money used to reduce a project's risk until private investors — pension funds, insurers, other institutional investors — will put in their own. Each source has a limit: public budgets compete with health and pensions, PPP tolls can make an asset costly to use, and development-bank loans add to national debt. Filling the gap is less about finding a single payer than matching each project to the mix of money it can realistically carry.

3.2.3. Vocabulary Focus

public-private partnership (PPP), infrastructure funding gap, multilateral development bank (MDB), concession, blended finance, institutional investor

3.2.4. Reading Questions

1. What misconception about infrastructure does the reading open by correcting?
2. What are the two forms the infrastructure funding gap can take?
3. In a public-private partnership, how does the private company earn money before it hands the asset back?
4. Why is a growing share of Japan's public-works spending going to maintenance rather than new construction?
5. Why does the reading describe blended finance as "public money used to reduce the risk"?

3.2.5. Discussion Questions

1. The reading lists four sources of infrastructure funding: public budgets, PPPs, development banks, and private investors. Which do you think should carry most of the burden, and why?
2. PPP toll roads shift the cost from taxpayers to the people who use the road. Is that fairer, or not? Explain.
3. Japan faces a large repair bill for infrastructure built decades ago. Should a government prioritise maintaining old assets or building new ones when money is short?
4. Compare the Nairobi Expressway model with fully public construction. Give two advantages and two risks of the PPP approach.
5. Design three conditions a government should attach to any PPP contract to protect the public interest.

3.2.6. Source Notes

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3.3. Trade Policies and Tariffs

3.3.1. Goal

Explain who bears the cost of a tariff and how tariffs and trade agreements pull an open economy in opposite directions.

3.3.2. Reading

Who actually pays a tariff? Legally the answer is clear: a tariff is a tax on imported goods, and the importer of record, the company bringing the goods across the border, pays it to customs, not the foreign seller^[1]. Economically it is less clear. The cost can split three ways — the exporter cuts its price, the importer accepts a thinner margin, or the final buyer pays more — and studies of the tariffs the US raised sharply in 2025 disagree on the split. One Federal Reserve analysis found about 90% of the cost fell on US firms and consumers^[1]; another found foreign exporters absorbed close to half through lower prices^[2]. This share, called the pass-through, depends on how much pricing power the exporter has.

Those 2025 tariffs were unusually large. The average effective tariff rate rose from about 2.4% in January 2025 to a peak near 28% in the spring, then settled around 17% by the autumn, the highest level since the 1930s^[3]. In February 2026 the US Supreme Court ruled the emergency law behind many of these tariffs did not authorise them; the government replaced them the same day under a different law, and the average rate fell to about 7% by mid-2026^[4].

Countries facing these tariffs responded in two very different ways. Japan negotiated: under a July 2025 framework, most US tariffs on Japanese goods were set at 15%, down from a threatened 25%, with cars cut from 25% to 15%. In exchange, Japan pledged up to US\$550 billion of investment in US industries such as semi-conductors and energy by 2029^[5].

India did not get a deal. In August 2025 the US placed a combined 50% tariff on Indian goods: a standard rate plus a 25% penalty tied to India's purchases of discounted Russian oil. It hit an estimated US\$48 billion of exports, and India's government called it "unfair, unjustified and unreasonable"^[6]. The same tool, a border tax on imports, was used in one case as a bargaining chip and in the other as a punishment.

Japan's exposure explains why it bargained hard. In 2025 the US became Japan's largest export market, taking about 18.6% of its exports, cars the single biggest category^[5]. A tariff is a barrier a government can raise at will; a free trade agreement does the opposite, a standing promise to keep barriers low. Japan belongs to one such agreement, the CPTPP, which since 2018 has removed tariffs on about 95% of goods traded among its members, well below the standard most-favoured-nation (MFN) tariff each would otherwise charge^[7]. None of the 2025 rates stayed fixed — levels moved repeatedly, and the US-China rate briefly rose above 100%^[3] — and for an economy sending nearly a fifth of its exports to one market, that unpredictability is itself a cost.

3.3.3. Vocabulary Focus

tariff, importer of record, pass-through, most-favoured-nation (MFN) tariff, free trade agreement, effective tariff rate

3.3.4. Reading Questions

1. Who is legally responsible for paying a tariff, and why does the reading say the economic cost is “less clear”?
2. By how much did the average US effective tariff rate change between January 2025 and the autumn of 2025?
3. What did Japan agree to give in exchange for lower US tariffs?
4. Why did the US place a combined 50% tariff on Indian goods?
5. What does the reading mean when it calls a free trade agreement “a standing promise to keep barriers low”?

3.3.5. Discussion Questions

1. The reading says a tariff’s cost is shared between exporter, importer and buyer. Who do you think ends up paying most in practice, and why?
2. The same tariff tool was used against Japan as a bargaining chip and against India as a punishment. Is it reasonable for a country to use trade taxes this way? Explain.
3. Japan sends nearly a fifth of its exports to the US. What are the risks of depending so heavily on one market, and what could Japan do to reduce them?
4. Compare a tariff with a free trade agreement as ways of managing trade. Which serves a mid-sized exporting economy better?
5. Advise a manufacturer that exports to the US on three ways to protect its business against sudden tariff changes.

3.3.6. Source Notes

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3.4. Pension Reform

3.4.1. Goal

Explain why ageing populations force governments to change public pension systems, and compare the main levers they use.

3.4.2. Reading

In April 2023, after months of strikes, France passed a law raising its minimum retirement age from 62 to 64, phased in by year of birth^[1]. It was meant to be complete by 2030. Instead, in late 2025, under budget pressure and renewed protests, France's parliament suspended the increase until after the 2027 election, freezing the minimum age at 62 years and 9 months^[1]. Even pausing a reform has a cost, and France's stop-start attempt shows how hard this lever is to pull.

Most public pensions are pay-as-you-go: today's workers' contributions pay today's retirees, with little capital saved in advance. A funded system instead invests contributions ahead of time, often in government bonds and shares. When retirees grow faster than workers, a pay-as-you-go system's sums stop balancing. Governments then have four levers: raise the retirement age, raise the contribution rate, cut the benefit formula, or bring more people into the system.

The United States acted early and then stalled. In 1983 it raised the statutory full retirement age from 65 to 67, a change still phasing in today^[2], and no major reform has followed. Its main retirement trust fund is now projected to run down in late 2032, after which payroll taxes would cover only about 78% of promised benefits unless Congress acts^[2].

South Korea pulled a different lever. In March 2025 its National Assembly approved the first increase in the pension contribution rate in nearly three decades, raising it in yearly steps from 9% of wages toward 13% by 2033^[3]. Raising contribu-

tions spreads the cost onto today's workers rather than delaying their retirement, but it is politically easier to pass than a higher pension age.

Japan has been adjusting continuously for 30 years. It legislated a rise in the pension age from 60 to 65 in the 1990s, only finishing the phase-in for men in 2025, and lifted the contribution rate to a fixed 18.3% of salary by 2017. Since 2004 an automatic mechanism, nicknamed the "macroeconomic slide," quietly trims benefit growth as the workforce shrinks. The replacement rate — pension income as a share of working-age income — is projected to fall from about 61% now toward a legal floor of 50% by 2057^[4]. Its most recent reform, in 2025, uses the fourth lever: widening coverage to draw an estimated 1.8 million more part-time workers into the system^[5].

The pressure behind all these changes is the same. That measure, the old-age dependency ratio, stood at 33 people aged 65 or older for every 100 of working age across the OECD in 2025, up from 22 in 2000, and is projected to reach 52 by 2050^[6]. The retirement age is the lever that matches that arithmetic most directly, and the one governments find hardest to hold in place, as France has just shown.

3.4.3. Vocabulary Focus

pay-as-you-go (PAYG), statutory retirement age, contribution rate, replacement rate, old-age dependency ratio, automatic adjustment mechanism

3.4.4. Reading Questions

1. What did France do to its retirement age in 2023, and what happened to that change in 2025?
2. What are the four levers governments can use when a pension system's finances stop balancing?
3. What is projected to happen to the main US retirement trust fund in late 2032?
4. Why does the reading say raising contributions is "politically easier" than raising the pension age?
5. What does Japan's "macroeconomic slide" do?

3.4.5. Discussion Questions

1. Of the four levers — higher retirement age, higher contributions, lower benefits, wider coverage — which is fairest for spreading the cost of an ageing population? Explain.
2. France raised its retirement age and then suspended the rise within two years. What does this suggest about how pension reform should be planned and explained to the public?
3. Japan uses an automatic mechanism that trims benefit growth without a new law each time. Is it a good idea to take these decisions out of politicians' hands? Give reasons.
4. Compare the US approach (one large reform in 1983, little since) with Japan's approach (continuous small adjustments). Which is better for a pension system's stability?
5. Advise a government whose pension fund is projected to run dry within ten years on a package of two or three measures it should take now, and say why.

3.4.6. Source Notes

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3.5. Global Economic Recovery Post-COVID

3.5.1. Goal

Use the 2020 pandemic downturn to show how governments respond to a global shock, and why recoveries from the same shock can diverge so sharply.

3.5.2. Reading

In the first months of 2020, the COVID-19 pandemic shut down travel, trade and daily commerce across most of the world at once, producing a recession that reached almost every country in the same quarter. Global output fell about 3.3% that year, a deeper one-year drop than the 2008 financial crisis^[1]. The fall was uneven from the start: the euro area shrank 6.8%, close to double the US decline, because lockdowns in parts of Europe were stricter and longer^[2]. What followed was not one recovery but several, at very different speeds.

Governments and central banks responded on a scale never seen in peacetime. Governments used fiscal stimulus — extra spending and tax relief running to trillions of dollars — with the US passing a large emergency package and the EU agreeing its first jointly financed recovery fund. Central banks used monetary easing, making borrowing cheaper: the European Central Bank's bond-buying programme, a form of quantitative easing (QE), grew from EUR 750 billion to EUR 1,850 billion^[3]. The shared principle was to spend whatever it took to stop a health emergency becoming a lasting depression, and in countries that could afford it, the feared collapse did not happen.

The support was deeply unequal. Rich economies could borrow cheaply and spend heavily; poorer ones spent far less as a share of their economies. By the World Bank's early-2026 assessment, roughly a quarter of developing economies were still poorer per person than in 2019, while almost every advanced econ-

omy had grown richer^[4]. Economists call this a K-shaped recovery, some parts rising while others keep falling, and the lost ground in the developing world is economic scarring — lasting damage a downturn leaves behind.

Even among wealthy countries the experience varied. The recovery brought a burst of inflation in 2022: US consumer prices rose 9.1% over a year, the fastest in four decades^[5], and the euro area's peak was higher still, above 10%. Japan's core inflation peaked much lower, at 4.2% in January 2023^[6] — its fastest pace since 1981, after 25 years of near-zero or falling prices, but still less than half the shock felt in Washington or Frankfurt. Japan also reopened its borders later and took longer to close its output gap, the gap between what an economy produces and what it could produce at full capacity: a slower cycle, not a sharper one.

The episode is now used as evidence that a fast, large response prevents the worst. But it left two unequal legacies: much higher public debt in the rich world, and lost ground in the developing world that no rebound has yet recovered. The open question for the next global shock is whether that speed and size can be repeated without again widening the gap between countries that can pay for a recovery and those that cannot.

3.5.3. Vocabulary Focus

fiscal stimulus, monetary easing, quantitative easing (QE), K-shaped recovery, scarring effect, output gap

3.5.4. Reading Questions

1. How did the size of the 2020 downturn compare with the 2008 financial crisis, and how did the euro area's fall compare with the United States'?
2. What was the shared principle behind the way governments and central banks responded in 2020?
3. Why did rich and poor countries end up with such different recoveries?
4. How did Japan's peak inflation compare with that of the United States and the euro area?
5. Why does the reading call the result "not one recovery but several"?

3.5.5. Discussion Questions

1. The reading says spending "whatever it took" worked in countries that could afford it. Was that the right approach, given the higher debt it left behind? Explain.
2. A global shock produced very unequal recoveries. Whose responsibility, if anyone's, is it to help the countries left behind?
3. Japan's inflation rose much less than in the US or Europe, yet the reading says it still "felt sharp." Why would a smaller price rise matter more in Japan than elsewhere?
4. Compare a fast, large policy response with a slower, more cautious one. What are the risks of each when the next global downturn comes?
5. Design three lessons from the COVID recovery that governments should build into a plan for the next economic shock.

3.5.6. Source Notes

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Part 4

Finance, Fairness and Global Cooperation

Finance, Fairness and Global Cooperation asks who bears the cost when the financial system is misused, and how far countries are willing to cooperate to stop it. Anti-money-laundering rules make banks the front line against criminals who try to disguise illegal money as legitimate, a system that only works if every country enforces it, since a gap in one jurisdiction can undermine strict rules everywhere else. Financial literacy programmes tackle a quieter version of the same fairness question: whether ordinary people have the knowledge to make sound financial decisions, or whether the system's complexity itself is a source of disadvantage.

Wealth inequality and corporate governance examine fairness from two more angles, one asking whether the very richest bear a fair share of the tax burden when their income is taxed differently from wages, the other asking whether a company's board genuinely represents shareholders or quietly serves management instead. Economic diplomacy closes the resource by taking the same fairness question to the level of nations, where sanctions and coordinated financial pressure are used as tools of foreign policy. Across all five topics, the resource ends where finance ultimately answers to something outside itself: not just markets, but law, ethics and international cooperation.

4.1. Anti-Money Laundering (AML) Regulations

4.1.1. Goal

Explain how governments try to keep the proceeds of crime out of the financial system by making banks the front line of detection.

4.1.2. Reading

At nine in the morning an analyst on a bank's compliance team opens a flagged payment: EUR 4 million moving from a company account opened three weeks ago to three firms abroad that were themselves registered only days earlier. Nothing is plainly illegal, but the pattern fits money laundering — moving the proceeds of crime through the financial system until the money looks legitimate when it comes out. The analyst must decide whether to release the payment, hold it, and file a suspicious-transaction report with the national financial-intelligence unit. Multiply that decision by millions a day and you have the front line of anti-money-laundering (AML) policy.

The problem is large but only roughly measured. The UN Office on Drugs and Crime still cites an old IMF estimate that 2–5% of world output is laundered each year, very roughly US\$800 billion to US\$2 trillion^[1]. Governments cannot watch every transaction themselves, so the law transfers the work to banks: identify each customer and the real person who controls a company, monitor for unusual activity, report what looks suspicious, and screen names against sanctions lists.

The global rulebook is set by the Financial Action Task Force (FATF), created by the G7 in 1989. Its standards shape law in more than 200 countries and territories, and it reviews each member through a peer “mutual evaluation,” publishing a “grey list” of countries under increased monitoring^[2]. A weak review raises the cost and friction of every cross-border payment a country's banks make, which

gives even reluctant governments a reason to comply.

Japan's system shows that pressure at work. The FATF's 2021 review found Japan's controls were “delivering results” but placed the country in “enhanced follow-up,” a status that requires regular progress reports. Tokyo responded with a burst of legislation, amending two laws to bring crypto-asset exchanges fully into the regime, including the “travel rule” that identifying information must accompany a transfer. Suspicious-transaction reports flow to JAFIC, a unit inside the National Police Agency, while the Financial Services Agency, Japan's main financial regulator, supervises the banks^[3].

The sharpest pressure, though, is financial. Denmark's largest bank, Danske Bank, let about EUR 200 billion of suspicious, mostly non-resident money pass through one small Estonian branch between 2007 and 2015; in December 2022 it pleaded guilty in the United States and forfeited about US\$2 billion^[4]. In October 2024 TD Bank became the largest US bank ever to plead guilty to Bank Secrecy Act failures, paying around US\$3 billion and accepting a legal cap on its US growth^[5]. After its own run of scandals, the European Union set up a central Anti-Money Laundering Authority in Frankfurt, which from 2028 will directly supervise around 40 of the highest-risk cross-border firms^[6]. Each scandal has been followed by a larger penalty or a new supervisor — an admission that the last line of defence was not enough.

4.1.3. Vocabulary Focus

money laundering; anti-money-laundering (AML); Financial Action Task Force (FATF); mutual evaluation; suspicious-transaction report; the “travel rule”

4.1.4. Reading Questions

1. What decision does the compliance analyst in the opening have to make?
2. Why does the reading say the scale of money laundering is “only roughly measured”?
3. What work does AML law transfer from governments to banks?
4. Why does a poor FATF review matter to a country’s banks, even though FATF cannot fine the country?
5. What did the reading mean by “each scandal has been followed by a larger penalty or a new supervisor”?

4.1.5. Discussion Questions

1. AML rules make private banks do the detection work. Is that the right body to carry it, or should governments do more themselves? Give reasons.
2. FATF has no power to fine a country, yet its reviews change behaviour. Why does that indirect pressure work?
3. Very large fines are meant to deter. Do you think a US\$3 billion penalty changes how a bank behaves, or is it treated as a cost of doing business?
4. Compare the FATF “grey list” and a record fine as tools for improving AML. Which would influence a bank more, and why?
5. You advise a mid-sized bank on where to spend a limited compliance budget: customer identity checks, transaction monitoring software, or sanctions screening. Rank the three and justify your order.

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4.2. Financial Literacy Programs

4.2.1. Goal

Weigh the evidence on whether organised financial-education programmes actually change how people manage money.

4.2.2. Reading

Does teaching people about money actually change what they do? For years the research said: barely. That answer is now contested, and the disagreement is the most useful thing about this topic.

Financial literacy programmes are organised efforts — school courses, adult workshops, national strategies, media campaigns — to raise people’s ability to make sound money decisions. Governments run or coordinate them because surveys keep finding large gaps in basic knowledge. The OECD’s 2023 survey of 39 countries and economies produced an average score of 60 out of 100, and in many countries only about a third of adults can correctly answer three simple questions on interest, inflation and spreading risk^[1].

The sceptical case came from a 2014 review of many earlier studies, which found that financial education explained only a fraction of a percent of later financial behaviour, and that whatever effect it had faded within about two years^[7]. That fed an argument that better default options and “just-in-time” guidance — help delivered exactly when a person faces a real decision — would beat classroom teaching.

The more recent evidence is more positive. A 2022 meta-analysis pooled 76 randomised experiments covering more than 160,000 people across 33 countries. It found financial education has real effects on both knowledge and behaviour — at least three times as large as the 2014 estimate, comparable to teaching in other subjects, with the strongest results for budgeting, saving and borrowing^[2]. The

honest summary is not “it works” or “it does not,” but that well-designed programmes delivered near a real decision show measurable effects, while poorly designed or badly timed ones show little.

Design and delivery show up in how countries act. In the United States, where the requirement is set state by state, the number of states guaranteeing every high-school student a standalone personal-finance course rose from a handful around 2020 to 25 by 2024^[3]. Earlier state mandates are found to improve young adults’ credit scores and cut their use of high-cost borrowing^[4]. The United Kingdom took a different route: its Money and Pensions Service runs a ten-year national strategy, launched in 2020, coordinating banks, charities and schools rather than delivering the teaching itself^[5].

Japan made financial education a priority for two concrete reasons. In April 2022 the age of majority dropped from 20 to 18, so first-year university students can now take out loans and credit cards without a parent’s consent, and revised school guidelines added household budgeting and “asset formation” to home-economics classes. Then in August 2024 the government launched a dedicated body, J-FLEC, jointly backed by the Financial Services Agency, Japan’s financial regulator, and the Bank of Japan^[6]. Its aim is to coordinate teaching as a form of consumer protection: education free of a seller’s incentive to push its own products. Whether a new agency and a school requirement change behaviour is exactly the open question the research leaves.

4.2.3. Vocabulary Focus

financial literacy; meta-analysis; randomised controlled trial (RCT); “just-in-time” education; national financial-education strategy; default option

4.2.4. Reading Questions

1. What did the 2014 review conclude about the effect of financial education?
2. How did the 2022 meta-analysis differ from the 2014 review in what it found?
3. What does the reading say separates programmes that show measurable effects from those that show little?
4. How does the United States decide whether high-school students get a personal-finance course?
5. What are the two concrete reasons Japan made financial education a priority?

4.2.5. Discussion Questions

1. The evidence on whether financial education “works” is genuinely mixed. Should governments keep funding it while the question is unsettled? Give reasons.
2. Some argue that better default options and rules do more good than teaching. Do you agree? Use an example.
3. The reading says education provided by a seller carries a conflict of interest. How serious is that problem, and how could it be avoided?
4. Compare the US state-mandate approach with the UK’s coordinated national strategy. Which looks more likely to change behaviour, and why?
5. Design a short financial-education programme for 18-year-olds in their first year at university. Say what it would cover, when it would be delivered, and how you would test whether it worked.

4.2.6. Source Notes

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2. Kaiser, Lusardi, Menkhoff & Urban: “Financial education affects financial knowledge and downstream behaviors,” *Journal of Financial Economics* 145(2) (2022) <https://www.sciencedirect.com/science/article/abs/pii/S0304405X21004281>
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4.3. Wealth Inequality and Redistribution Policies

4.3.1. Goal

Explain how wealth inequality is measured and which redistribution tools actually move the numbers.

4.3.2. Reading

The richest 10% of the world's adults own about 76% of all household wealth. The poorest half own about 2%^[1]. The gap in what people own is far wider than the gap in what they earn, because wealth compounds over time and passes between generations, while a wage stops when the work does.

Inequality is usually summed up in one number, the Gini coefficient, which runs from 0 (everyone equal) to 1 (one person owns everything). In the United States, Federal Reserve data show the wealthiest 1% of households owned about 30.9% of roughly US\$162 trillion in household wealth in 2024, while the bottom half owned about 2.5%, down from around 3.4% in 1989^[2]. Across the OECD, taxes and cash transfers narrow the income gap — by more in high-transfer systems such as France and the Nordic countries, and by less in the United States and South Korea.

The redistribution toolkit has four main parts: progressive income tax, which takes a larger share as income rises; transfer payments such as pensions and child benefit; taxes on capital gains and inheritance; and, more rarely, an annual tax on a person's total assets. That last tool has mostly been tried and dropped. The number of OECD countries levying an annual net wealth tax fell from 12 in 1990 to just three by 2020 — Norway, Spain and Switzerland^[3]. The common reasons: the taxes raised little, were hard to value and enforce, and prompted wealthy residents to move assets abroad or leave.

Japan shows that a tax system's design matters more than its headline rates. Its

income inequality is moderate by rich-country standards, but its effective tax rate actually falls for the very highest earners: income from shares or dividends is taxed at a flat 20%, while employment income is taxed progressively up to about 45%. Since the wealthiest take more of their income as investment returns, the average rate can start dropping past about JPY 100 million — the “JPY 100 million wall.” The government's answer, phased in from 2025, is a minimum tax of at least 22.5% on income above JPY 330 million, tightening further from 2027^[4].

The same problem is now on the international agenda. Brazil, one of the world's most unequal large economies, used its 2024 G20 presidency to commission a blueprint from economist Gabriel Zucman for a coordinated minimum tax on the ultra-rich: at least 2% of wealth each year from individuals worth over US\$1 billion. Applied to roughly 3,000 known billionaires, that would raise an estimated US\$200–250 billion, against an effective rate today of about 0.3%^[5]. In July 2024 G20 finance ministers issued a first joint statement on taxing the very wealthy, non-binding, with the US and Germany among the holdouts^[6]. Taxing wealth is less about rates than about which kinds of income and assets are taxed lightly — and the wealthiest hold exactly those.

4.3.3. Vocabulary Focus

wealth (net worth); Gini coefficient; progressive / regressive tax; transfer payment; net wealth tax; effective tax rate

4.3.4. Reading Questions

1. Why is the gap in what people own wider than the gap in what they earn?
2. What does a Gini coefficient of 0 mean, and what does 1 mean?
3. What happened to the number of OECD countries with an annual net wealth tax between 1990 and 2020, and why?
4. Why can the average tax rate fall for Japan's highest earners?
5. What would the Zucman blueprint require of the world's billionaires, and how much would it raise?

4.3.5. Discussion Questions

1. Annual wealth taxes have mostly been repealed. Does that mean they are a bad idea, or that they were badly designed? Explain.
2. The reading says tax design matters more than headline rates. Do you agree? Use the Japanese example.
3. A global minimum tax on billionaires needs many countries to act together. Why is that so hard, and is it achievable?
4. Compare taxing income and taxing wealth. Which is easier for a government to do well, and why?
5. You advise a government that wants to reduce wealth inequality without driving wealthy residents abroad. Recommend two measures from the toolkit and explain why you chose them over the others.

4.3.6. Source Notes

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<https://wir2022.wid.world/executive-summary/>
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4.4. Corporate Governance and Accountability

4.4.1. Goal

Trace how corporate-governance rules developed — mostly after scandals — and explain the chain of accountability they try to enforce.

4.4.2. Reading

It is often said that shareholders run a company they own, and that a board simply works for the chief executive. Both are wrong. Shareholders elect a board; the board is supposed to hire, monitor, pay and, if necessary, fire the management on the owners' behalf. Corporate governance is the set of rules and checks that tries to make that chain of accountability actually work, developed mostly in response to failures.

The first modern code was Britain's Cadbury Report of 1992, written after a run of company collapses and frauds. It recommended separating the roles of chair and chief executive, using outside non-executive directors, and setting up audit committees. Its lasting idea was “comply or explain”: a company may depart from the code, but must publicly say why^[1]. The United States went further after Enron collapsed in 2001. The 2002 Sarbanes-Oxley Act made governance duties statutory, requiring executives to personally certify the accounts, and created a new body to police the auditors^[7].

The pattern — scandal, then rules — repeated elsewhere. In 2009 the founder of the Indian IT firm Satyam confessed to inflating its accounts for years, feeding directly into India's Companies Act 2013, which required listed boards to include independent directors^[8]. In Germany, Wirecard, a DAX-listed payments company, collapsed in June 2020 after admitting that about EUR 1.9 billion of its cash had probably never existed. Its auditor had signed off for a decade; the regulator, meanwhile, had spent its energy investi-

gating the journalists who raised the alarm^[2].

Japan has since run a sustained governance-reform programme. Before 2014 many listed Japanese firms had few independent directors and held blocks of each other's shares to cement business ties, not to earn a return. A Stewardship Code in 2014 and Japan's first Corporate Governance Code in 2015, both on “comply or explain,” pushed companies to add outside directors, disclose more, and unwind those cross-shareholdings^[3]. In 2023 the Tokyo Stock Exchange went further, publicly pressing the roughly half of its top-tier “Prime” companies trading below the value of their own net assets to publish plans to improve capital efficiency^[4]. The reforms are credited as one trigger for the strong 2023–24 rise in Japanese share prices.

Governance also works from the other direction, when shareholders act through the board. In 2021 a tiny activist fund, Engine No. 1, holding about 0.02% of ExxonMobil, won three of twelve board seats by persuading large institutional investors — index funds and public pension funds — to back its candidates^[5]. It was the first board election decided on climate strategy, and a concrete demonstration that the owners-elect-the-board mechanism can change a company's direction^[6]. Three decades of codes and laws have not prevented every scandal; what they have built is a chain of accountability that holds when someone chooses to pull on it.

4.4.3. Vocabulary Focus

corporate governance; board of directors; non-executive (independent) director; “comply or explain”; cross-shareholding; shareholder activism

4.4.4. Reading Questions

1. What is wrong with the idea that “shareholders run the company”?
2. What does “comply or explain” allow a company to do?
3. In the Wirecard case, which checks failed, and what had the regulator been doing instead?
4. What three changes did Japan’s 2014 and 2015 codes push companies to make?
5. How did a fund owning only 0.02% of ExxonMobil win three board seats?

4.4.5. Discussion Questions

1. Governance rules have mostly followed scandals rather than prevented them. Is that inevitable, or could rules be more forward-looking?
2. The UK uses “comply or explain” codes; the US uses hard law. What are the strengths and weaknesses of each approach?
3. Japan’s reforms are credited with helping lift share prices, but critics say disclosing a plan is not the same as delivering results. How would you tell whether the reforms worked?
4. Should a fund owning a tiny stake be able to reshape a large company’s board? Give reasons for and against.
5. Compare the Cadbury Report (1992), the Wirecard collapse (2020) and the Exxon board vote (2021) as moments in this history. Put them in order of how much each changed corporate governance, and defend your ranking.

4.4.6. Source Notes

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4.5. Economic Diplomacy and International Cooperation

4.5.1. Goal

Show how governments use financial and trade ties as foreign-policy tools, and why doing so is costly to the country that tries it.

4.5.2. Reading

Economic diplomacy is the use of trade, investment, aid and financial ties — including the threat of cutting them off — as tools of foreign policy. It runs on two tracks: coercion, such as sanctions and asset freezes, and cooperation, such as coordinated crisis lending.

The clearest recent case of coercion is the response to Russia's invasion of Ukraine in February 2022. A coalition of all G7 economies and the EU froze roughly US\$300 billion of the Russian central bank's reserves, cut major Russian banks out of SWIFT, and capped Russian sea-borne oil at US\$60 a barrel^[1]. It was the largest coordinated use of financial sanctions ever applied to a major economy. About EUR 200 billion of the frozen reserves sit at a single securities depository in Belgium; rather than seize the principal, the EU redirects the profits to Ukraine, and in 2024 the G7 agreed a roughly US\$50 billion loan, repayable from that future revenue^[1].

The episode works because deep financial connections are themselves leverage, concentrated in the US dollar. The dollar dominates reserves, trade invoicing and cross-border payments, so the United States can impose “secondary sanctions” — penalising firms in third countries for dealing with a sanctioned party. The cooperative track uses the same connections for shared problems: the G20 was raised to leaders' level during the 2008 financial crisis, and it later worked with the IMF and World Bank to restructure the debts of the poorest countries.

Climate finance is the clearest current example of that cooperative track, and of its limits. Rich countries pledged at Copenhagen in 2009 to mobilise US\$100 billion a year for developing countries by 2020, a target met only around 2022^[2]. At COP29 in Baku in November 2024 they agreed a new goal — at least US\$300 billion a year by 2035, within a broader aspiration of US\$1.3 trillion — which many developing countries called far too low^[3].

Japan's handling of the 2022 sanctions shows the balancing act involved. It joined the G7 measures despite importing almost all its oil and gas, but drew one line. The trading houses Mitsui and Mitsubishi kept their 12.5% and 10% stakes in Sakhalin-2, a gas project supplying about 9% of Japan's LNG^[4], and Tokyo secured a G7 carve-out exempting it from the price cap^[5]. The reasoning was blunt: solidarity with allies matters, but a country that cannot fuel its power stations has no foreign policy at all.

Every use of leverage has a cost to the side applying it. Europe absorbed a severe energy-price and inflation shock in 2022, and Russian oil was simply re-routed, often sold above the cap. Targets also gain a reason to build alternatives, as China has with the Asian Infrastructure Investment Bank and a growing yuan-based payments network offered as an alternative to dollar clearing^[6]. The 2022 sanctions imposed large costs without forcing a change of course — which is why economic diplomacy is best described as leverage, not control.

4.5.3. Vocabulary Focus

economic diplomacy; sanctions; asset freeze; SWIFT; secondary sanctions; price cap

4.5.4. Reading Questions

1. What are the two “tracks” of economic diplomacy described in the reading?
2. What three main measures did the G7 and EU take against Russia in 2022?
3. Why does the reading say the leverage behind sanctions is “concentrated in the US dollar”?
4. What line did Japan draw when it joined the 2022 sanctions, and why?
5. What costs does the reading say fall on the country that applies sanctions?

4.5.5. Discussion Questions

1. The reading calls economic diplomacy “leverage, not control.” What is the difference, and why does it matter for policy?
2. The EU redirected the earnings on Russia’s frozen reserves rather than seizing the reserves themselves. Why might it have stopped short of seizure?
3. Sanctions push targets to build alternative institutions and payment systems. Is that an acceptable price, or a reason to use sanctions less?
4. Compare the coercive and cooperative tracks. Which do you think does more to advance a country’s foreign-policy aims?
5. You advise a government deciding whether to join an allied sanctions coalition against a country it depends on for energy. Set out the case for joining, the case against, and the conditions you would attach.

4.5.6. Source Notes

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